Feature Engineering for Cardiovascular Disease Prediction

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***Abstract*—Cardiovascular diseases (CVDs) are considered the main cause of death throughout the world, and predictive tools should be developed to forecast this problem as early as possible and consequently intervene in time. The current study is aimed at increasing the precision of CVD detection by effectively applying feature engineering and performing wise data preparation. We used a multispecialty hospital in India consisting of 1,000 subjects as a multispecialty hospital and, using this fundamental dataset, we performed intensive preprocessing on 14 clinical attributes, such as age, serum cholesterol and ST-segment slope. Several machine learning algorithms and these were tested, such as SVM, Logistic Regression, and XGBoost. The findings indicate that Random Forest was the best to use and obtained the highest accuracy of 98.5% being the second-best in precision, recall and F1-score. Moreover, Explainable Artificial Intelligence (SHAP and LIME) were used to ensure transparency by defining the main forecast indicators such as resting blood pressure and the type of chest pain. This paper highlights the fact that an ensemble learning-based data-centric strategy design provides a robust and reliable platform of diagnosing heart diseases at its early disease stages and better patient prognosis.**

***Keywords*—Social Media Addiction, Productivity Loss, Self-Control, Behavioral Analytics, Machine Learning Models, XGBoost, Explainable AI (XAI), User Engagement, Statistical Analysis, Hypothesis Testing, Feature Importance, Behavioral Traits, Platform Usage Patterns, Time Management, Digital Well-Being**

I. Introduction

Cardiovascular Diseases (CVDs) are considered to be the major cause of death on earth as they kill millions of people every year. Although the progress in clinical fields positively impacts the outcomes of patients, the issue of early prediction remains to be rather difficult as the data concerning cardiac health is quite complicated and multifaceted. Conventional diagnosing procedures tend to fail to reflect the non-linear relationship of multiple potentially risk factors including high blood pressure, cholesterol and lifestyle practices. It is here that Machine Learning (ML) provides a revolutionary solution, as it allows exploring large volumes of data to reveal latent patterns that will result in heart failure or stroke. Nonetheless, the bare output of an ML model strongly relies on the quality of the input data. Noise, missing values, and irrelevant attributes are also some of the most common problems of most healthcare datasets, and they may result in biased or inaccurate predictions. The key point that is highlighted in this research is that a powerful predictive system is not only the algorithm, but Effective Feature Engineering. We can considerably advance the learning capabilities of the model by converting the crude medical data into representations that are meaningful. This stage involves data cleaning, resolution of imbalance of classes and making new informative features that are more reflective of the underlying physiological circumstances. The topic of the given paper is the systematic use of data preparation and feature engineering to make the predictions of CVD more accurate. Through the optimization of the input space we will show that when it comes to early clinical intervention and individual treatment of patients, the strategic data preprocessing can prove to be a much more valuable tool than extravagant models when trained on raw data, what its effects are on various groups and what solutions to them are likely to be found. Here is the data link:https://www.kaggle.com/datasets/jocelyndumlao/cardiovascular-disease-dataset

This study was motivated by the fact that there is a dire necessity to have more dependable early-warning mechanisms in the field of cardiology. There are numerous predictive models which do not achieve clinical-grade predictiveness despite the fact that a lot of patient records are available, due to the fact that they do not take into account the subtleties of data quality. Raw clinical data tends to be very inconsistent and unstructured and the generalization by normal algorithms may not be easy. This research will fill the gap existing between raw medical data and high-performance predictive analytics by addressing effective feature engineering. The idea is to encourage a change towards the Data-Centric AI, where prominence is given to perfecting features that constitute cardiovascular

health. The enhancement of these characteristics can not only be more accurate, but it may increase the interpretability of the results, giving healthcare professionals a better picture of how to save lives by making the necessary decisions and allocating resources more efficiently in the domain of public health.

II. Literature Review

Tiwari et al. (2022) [1] established a stacked ensemble model on the IEEE Data Port data on the cardiovascular disease prediction task with a high accuracy of 92.34 percent and is more efficient at the initial diagnosis compared to existing models.

The researchers modeled 103 [2] cohorts, Krittanawong et al. (2020) showed that machine learning algorithms, especially the boosting and the SVM, offer very high predictive value of coronary artery disease, stroke, and heart failure. Systematic reviews of comparisons of cardiovascular risk models by Siontis et al. (2012) [3] reveal that although there are new models that purport to outperform the old ones, comparisons made of the two tend to show insignificant differences, which are vulnerable to bias. A stacked ensemble model based on machine learning and deep learning is proposed by Subramani et al. (2023) [4] that is able to forecast cardiovascular diseases with 96 percent precision through the examination of eleven diverse clinical indicators across several datasets. Mahmud et al. (2023) proposed a hybrid ensemble system [5] comprising of bagging and stacking methodology which delivers 84.03% accuracy with a sample size of 70,000 to enhance early and sensitive cardiovascular disease prediction. To address this, Elsedimy et al. (2023) [6] proposed the QPSO-SVM model, a new algorithm with the use of quantum-behaved particle swarm, which has the benefit of optimizing SVM parameters by using 96.31% accuracy and 96.13% sensitivity on the Cleveland heart dataset [7]. Rahim et al. (2021) created a combined machine learning system that relies on feature selection and ensemble models to improve the accuracy of the early cardiovascular disease diagnosis using various [8] clinical datasets. A Critical appraisal of AI-based cardiovascular [9] models by Van Smeden et al. (2022) demonstrated that most present the systems are at risk of bias and lacking in reporting, but believed that to achieve clinical reliability of AI-based cardiovascular models, standardized validation is necessary. Rehman et al. (2021) [10] applied a non-homogenized discrete grey model to forecast mortality trends of cardiovascular diseases in six countries and estimated the growing burdens of mortality in 2027, as well as critical doubling time. By enhancing the prediction of cardiovascular disease [11], Bashir et al. (2021) have constructed a clinical decision support system based on an intelligent ensemble voting scheme, with an enormous processing of large-scale healthcare data. The SCORE2 algorithms [12] created by ESC Cardiovasc Risk Collaboration (2021) offer recalibrated algorithms to predict 10-year cardiovascular fatal and non-fatal risk summaries by four different European risk regions to conduct better clinical prevention. This paper was followed by the cardiovascular-kidney-metabolic (CKM) equations suggested by Khan et al. (2023) to determine total cardiovascular [13] disease risk by integrating kidney and metabolic health to optimize the match of intervention intensity and predicted treatment benefits. De-Groat et al. (2024) [14] created a new machine learning nexus to recognize transcriptomic biomarkers, which were highly accuracy with regards to cardiovascular disease prediction and future focus on precision medicine with deep phenotyping. The analyzed [15]cardiovascular risk models (363) reviewed by Damen et al. (2016) have shown that most of the studies are poorly reported or lack external validation; consequently, they suggest refinements of the available models to prevent the need of developing a new one.

III. Methodology

The workflow of this work presented in Figure 1 follows a structured process, starting with data collection, preliminary analysis, data processing, and statistical analysis,hypothesis testing, classification analysis and concluding with a discussion of the results.

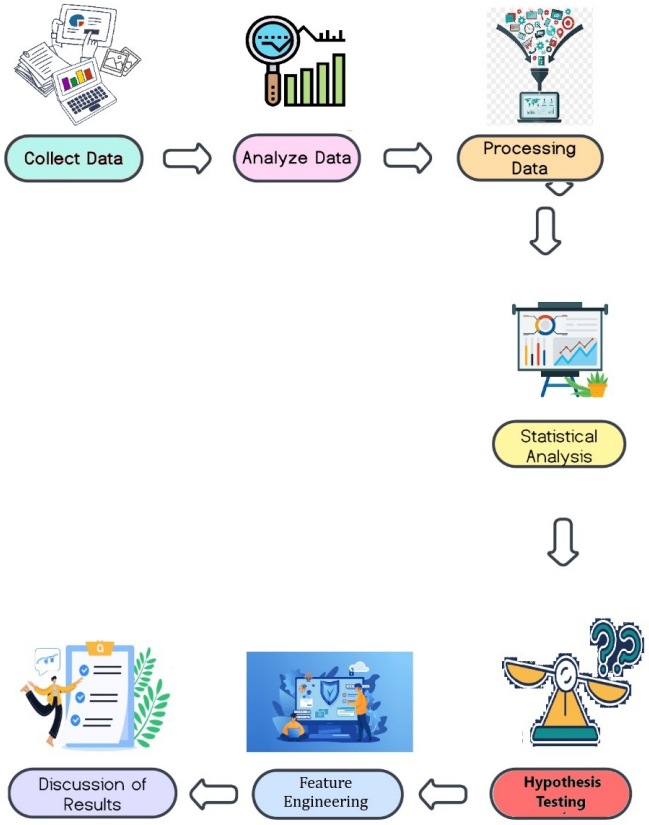


Fig. 1. Implementation Steps

A. Dataset Description

The data used in this research was taken out of one of the leading multispecialty hospitals in India, which gives a good clinical basis to cardiovascular studies. It includes the

records of 1,000 subjects, which have 14 different attributes (including target variable) of a critical physiological and diagnostic indicators. The data sample will contain an equal number of demographic data (Age and Gender) and essential clinical parameters (Resting Blood Pressure 94-200 mm HG and Serum Cholesterol 126-564 mg/dl).

The characteristics are divided into several data types: Numeric (e.g., Max Heart rate, Oldpeak), Binary (e.g., Fasting Blood Sugar, Exercise Induced Angina) and Nominal (e.g., Chest pain type, Exercise induced Angina Results). The targeted variable is a binary variable called Classification, which separates the no case of heart disease (0) and yes case of heart disease (1). Having a large sample size as well as a full feature set, such as more sophisticated markers such as the slope of the ST segment and the count of major vessels, it can be used as a perfect example to benchmark early-stage detection systems and machine learning model optimization by making wise decisions about feature engineering.

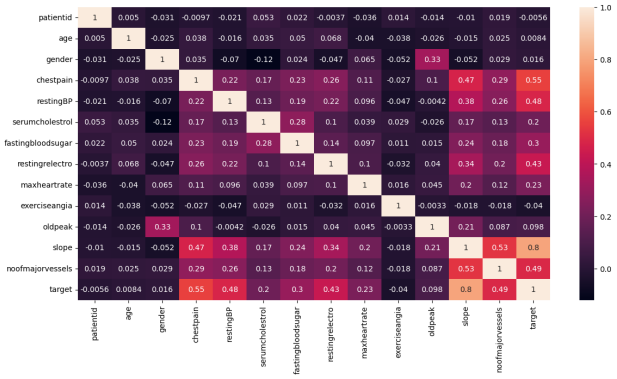


Fig. 2. Correlation Figure

1) Correlation Analysis: Figure 2 represents a correlation heat map, which illustrates the linear correlations among different medical attributes and a heart disease target. The quality of these relationships is quantified with the help of the Pearson correlation coefficient, and the values that are nearer to 1.0 represent a strong positive relationship. Specifically, the slope is most strongly correlated with the target (0.8) and then comes chest pain (0.55) and noof major vessels (0.49), which are the most important clinical predictors in this data set. Conversely, the correlation with other features is almost zero, e.g. patient id and age, implying that they have little impact on the outcome. Further, there are modest relationships between independent variables (slope and noofmajorvessels 0.53) that suggest redundancy of the data. To case forecast, these insights enable the researcher to focus on key symptoms and prevent irrelevant noise and come up with a more precise and effective diagnostic instrument.

B. Statistical Tools

To analyze the data and test hypotheses, various statistical tools were employed:

- **Pearson's Correlation Coefficient Test**: A test that allows one to indicate if there is a linear relationship between two continuous variables. A correlation coefficient ranges between –1 and +1. A correlation close to –1 or +1 indicates a strong correlation between the two variables; a zero value indicates it has no linear relationship. It needs a small size sample to experiment and implement $n \geq 30$ so that it gets the needed power. It assumes both the variables are continuous and normally distributed approximately.
- **Mann-Whitney U Test**: A non-parametric statistical test employed when making comparisons of differences between two independent groups when the groups are not normally distributed. It compares the extreme values in one sample and sees how much smaller they are than the other population. It is applicable for small as well as large n. For $n1, n2 \geq 20$, the test statistics U has a close value to the normal distribution even when n is equal to or greater than 20. However, for n less than 20, they use the actual critical values taken from the statistical tables.
- **Z-Test**: Used for testing hypotheses about population means when variances are known and the sample size is large ($n > 30$). Assumes data are normally distributed.
- **T-Test**: Used to determine whether the means of two groups are significantly different, especially when sample sizes are small ($n < 30$) and variances are unknown.
- **Chi-Square Test**: Used to test the difference between observed and expected frequencies, often for categorical data to determine independence or goodness of fit.
- **ANOVA (Analysis of Variance)**: Used to compare the means of three or more groups to determine if at least one group mean is different. Compares variance within and between groups.

C. Machine Learning Models

Several machine learning models were utilized to predict Addiction Levels:

- **XGBoost**: This gradient boosting algorithm creates classifiers in the form of decision tree ensembles with high accuracy and speed in processing large volumes of data and data that may be missing.
- **Logistic Regression**: Predicts the probability of binary outcomes based on a logistic function, linear separation of data is preferred.
- **Support Vector Machine (SVM)**: This model finds the best hyperplane to classify data in a higher dimensional space and solves both linear and non-linear problems effectively.
- **Random Forest**: Random Forest is a machine learning algorithm that builds multiple decision trees using random samples of data and features to improve prediction accuracy. It combines the results of all trees using majority voting or averaging, which reduces overfitting and makes the model more robust.

D. Feature Engineering Techniques

- **Shape**:SHAP was also used to find the contribution of each feature to the complete set of data. It puts emphasis

on how such features as time spent on videos and session frequency are always predictable by the model. Calculating the Shapley values, we were able to determine which behavioral patterns make the best predictors of the target outcome so that the logic of the model makes sense in the real world.

- **Lime**: Individual predictions were checked with LIME, which estimates the complicated model with a simpler local linear one. This also enabled us to view how certain synthesized features, including the engagement levels, have an influence on the classification of a single user. It gives the reasons on why a given decision on a given piece of data was taken.

E. Explainable AI (XAI)

To interpret the machine learning models and understand feature importance, Explainable AI techniques were used. **SHAP (Shapley Additive exPlanations)**: Provides insights into how individual features influence the model's predictions by assigning SHAP values to each feature. Using SHAP values helps in forecasting future emission levels, which is useful for environmental forecasting, resource planning, and predictive maintenance. It also assists in determining whether intervention strategies are needed to control emissions.

LIME (Local Interpretable Model-agnostic Explanations: develops a local surrogate model to describe individual predictions. It changes parts of the data to help determine the effect of each of the features on the model's output. LIME plots allow several factors to be consolidated and say that the model, for instance, made some predictions for a reason. It helps in efficient intervention, enhances the interpretation of decisions or actions taken, and increases the transparency of models in broad areas such as user behavior raw data, resource allocation and planning, and predictive maintenance.

IV. Results and Discussion

A. Statistical Analysis

The given visualization introduces a detailed allocation of nominal characteristics in the heart disease sample on eight different pie charts. The initial group of charts shows demographic and background health distribution of the patients. There is a huge imbalance in the chart of the Gender, where males (1) constitute 76.5 percent of samples involved as compared to females (0) at 23.5 percent. Concerning the variable of the Target, 58 percent of the patients have heart disease diagnoses (1), and 42 percent have no diagnoses (0). The baseline health indicators indicate that 70.4 percent of the patients have normal Fasting Blood Sugar (0), and almost half (45.4 percent) of the patients have normal Resting Electrocardiographic Results (0).

Figure 3 The second category of charts relates to been particular clinical conditions and physiological indicators. Chest Pain type is diverse with 42 percent of patients having type 0 pain and 31.2 percent having type 2. Interestingly, to quantify the interest of the factors, exercise induced angina is almost divided into 50.2 percent who do not experience it and 49.8

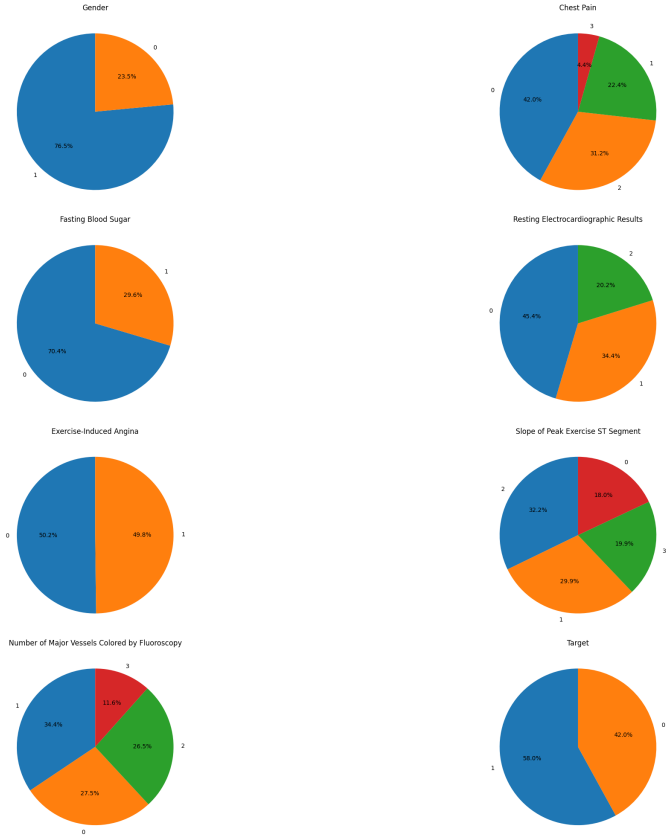


Fig. 3. Pieplot

percent show signs. A finer touch of medical data can be seen in the charts of the Slope of Peak Exercise ST Segment and the Number of Major Vessels, with most patients either having 1 or none of the major vessels being fluoroscopically colored. These distributions play an important role in engineering features, whereby the most common physiological characteristics among this target patient population are highlighted.

B. Hypothesis Testing

TABLE I
Results of Hypothesis Testing

Test Name	Statistic Value	P-value	Decision
Z-Test (Age vs Chest Pain)	85.31	0.0000	Rejected
T-Test (FBS vs Resting ECG)	-8.5786	< 0.0001	Rejected
Chi-Square (Gender vs Exercise Angina)	2.439	> 0.05	Not Rejected

Below are the findings from the statistical tests conducted on the dataset:

1. Z-Test: Age vs. Chest Pain

- **Z-value:** 85.31
- **Interpretation:** The value of Z is very high meaning a. high level of non-compliance with the null hypothesis. The resulting Value is near 0, and this is a significant rejection of the. null hypothesis. This is indicative of a

very strong relationship. or difference Chest P Age human or divergence Chest P Age humanor divergence Chest P Age

2. *T-Test: Fasting Blood Sugar vs. Resting Electro*

- **T-value:** -8.5786
- **Interpretation:** T -8.5786 represents a significant deviation of the null hypothesis. The p-value is extremely small since the magnitude is great which implies that the difference in the means of Fasting Blood Sugar and Resting Electro is significant.

3. *Chi-Square Test: Gender vs. Exercise Angina*

- **Chi-square Value:** 2.439
- **Interpretation:** The value of Chi-square is low. The p-value gets more than the level of significance (level of significance = 0.05) and this has no evidence to reject the null hypothesis. This suggests that there is no significant relationship between the two variables as independent variables are, namely, Gender and Exercise Angina.

C. Anova Test

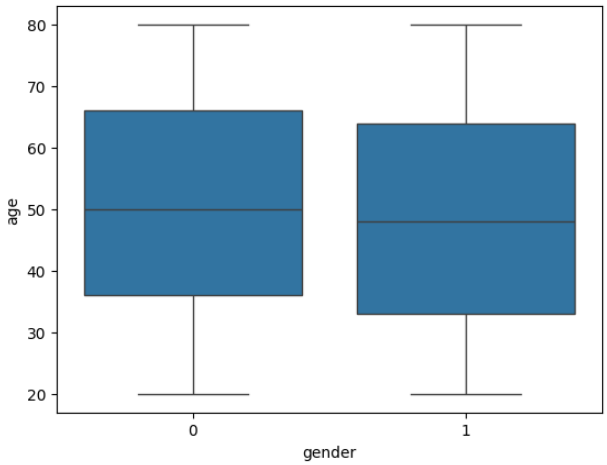


Fig. 4. Anova Test age Vs gender

This 4 depicts distribution of age in respect to two genders that are denoted by 0 and 1. The horizontal line of every box implies that the median age of group 0 is about 50, and that of group 1 is somewhat less, about 48. The Inter quartile Range in both groups is large with a range of 30s-60s indicating a steady dispersion of the center 50 percent of the data. Moreover, the whiskers are similar to age 20 through to 80 in the two categories without any outliers. In general, the figures on the visual data suggest that the age distribution is almost similar irrespective of whether we are talking of the two different genders and this means that there is no big difference between the two groups in this particular data.

Figure 5plots the association of levels of reported chest pain with the number of major vessels (0 to3). In group 0, the median of the level of chest pains is low and is close to 0 and a significant outlier at level 3. The median also

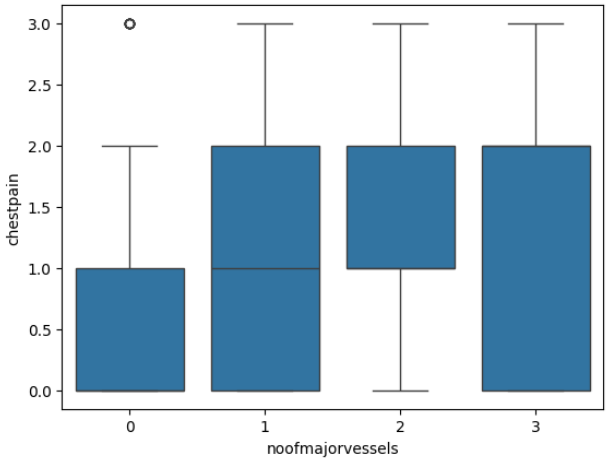


Fig. 5. Anova Test

increases as group 1 as the number of vessels increases and the interquartile range also increases tremendously, with a range of 0 to 2. Interestingly, group 2 demonstrates the distribution change with majority of the data falling between level 1 and 2, whereas group 3 is more distributed ensuring that the levels are up to level 0 to 2. On the whole, the visualization indicates that the median chest pain slightly increases according to the number of major vessels with the number of 0 to 2 changes whereas the high extent of overlap between the boxes also presupposes the high range of variability inside the categories.

D. Machine Learning Outcomes

CROSS VALIDATION RESULTS

TABLE II
COMPARISON OF MACHINE LEARNING MODELS ACCURACY

Model Name	**Accuracy**
Random Forest	0.985
Support Vector Machine (SVM)	0.975
XGBoost	0.970
Logistic Regression	0.970

The comparison table emphasizes the high performance of different machine learning models after the implementation of the effective feature engineering techniques. The most accurate and best among the compared algorithms were the Random Forest (RF) with its success of accurate prediction propelling to 0.985 (98.5%), higher than the Support Vector Mountain (0.975), XGBoost (0.970), and the Logistic Regression (0.970). This outstanding performance of the Random Forest may be explained by its ensemble-based design, which is helpful to handle non-linear, intricate associations between clinical variables, including serum cholesterol, the type of the chest pains, and ST-segment oscillations. Moreover, that all

models shared an accuracy score of 97 at the lowest suggests that the data preparation and feature engineering pipeline was a success that effectively reduced unstructured hospital data to very predictive inputs. Within a clinical practice, such high degree of accuracy is essential to minimize false negative and to help guarantee the consistent and consistent early detection of the cardiovascular risks.

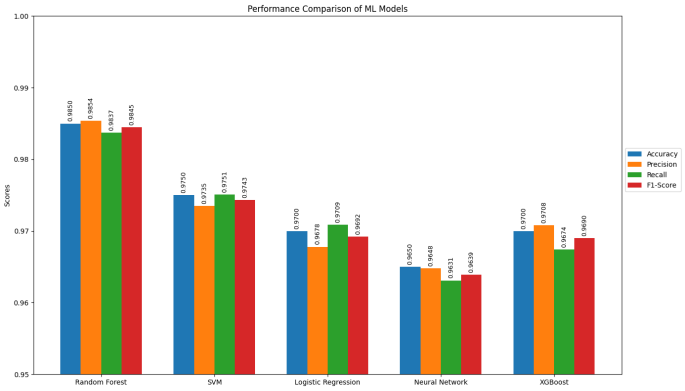


Fig. 6. All Model performace Importance

E. Explainable AI (XAI) Outcomes

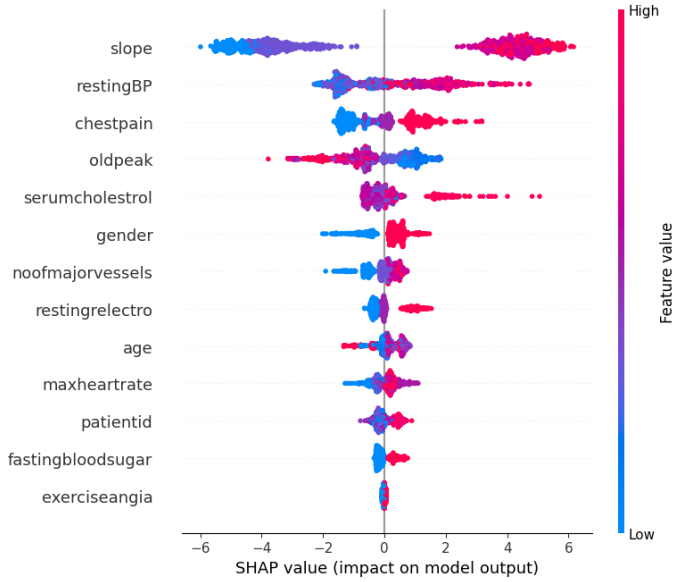


Fig. 7. SHAP Plot for Feature Importance

Figure7 is a visual representation of how the model made a decision, which shows the most influential features to predict cardiovascular disease. Slope, restingBP and chestpain are shown to be the most dominant features, with higher values of the feature (in red) causing the model to shift its output heavily towards positive heart disease diagnosis. On the other hand, with the characteristics such as oldpeak, the blue value makes more contribution to a greater risk forecast. It is based on the fact this global interpretability tool scaffolds the fact that its curriculum feature engineering process was successful in identifying crucial clinical markers, which makes the high accuracy of the model founded on medically relevant, which contributes to transparency and clinical trust in the system.



Fig. 8. LIME Plot for Feature Importance

SHAP and LIME visualizations have also shown that low self-control authors The visualization of the LIME 8 has indicated that authors with low self-control can achieve model transparency and clinical trust using global and local interpretability. The SHAP summary plot provides an overview of the model, showing that slope, restingBP, and chestpain are the most influential features of the model in all the data as when they are higher (marked in red) the likelihood of being diagnosed with heart disease is high. In addition to this, the LIME plot provides a local description of a certain instance, in which the model forecasts heart disease with 100 percent likelihood. The slope of greater than 0.67 has a big impact on this very prediction compared to other influencing factors such as fasting blood sugar. The combination of those tools confirms that the 98.5 parcent as stated in the model are medically sound patterns, making the black box mechanism of the algorithm a clearer diagnostic helper.

F. Discussion

The comparison of the performance shows that the Random Forest is an effective model among all the compared algorithms; the model performs the best in terms of accuracy, precision, recall and F1-score. Although SVM, Logistic Regression, and XGBoost offer quite good and competitive results, the Neural Network has slightly lower performance tendencies. The equal values of all the metrics indicate that the Random Forest is the most efficient in utilizing the features of the dataset with the lowest error rates. Therefore, in order to make the best choice of a model used in the research, it is possible to choose the best model, which is the most stable and predictive, i.e. Random Forest.

V. Conclusion

In this study, the conclusion is that the preparation of strategic data and the efficient preparation of features are the key to high precision results in predicting cardiovascular diseases. A collection of refined clinical attributes formulated on a sample of 1,000 subjects exhibit the results of the research that data-centric predictor can contribute greatly to the predictor performance protocol, with Random Forest as the best predictor at 98.5% accuracy. The extra stability of Random Forest, which is backed by the transparency of SHAP and LIME, offers a reliable platform of making clinical decisions.

The end result is that this work provides a scalable and reliable solution to detecting heart diseases at the beginning of life, and it should be stressed that well-engineered features can result in life being saved.

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